AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 10 · STUDENT LESSON

Image Generation Models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Inspect the process. Check the picture. Bound the claim.

Explain a supplied image-generation illustration while separating visual criteria and event authenticity.

Keep one habit: Origin contract → visible poster → complete grid trace → supplied process cards → independent audit.

MAIN LESSON · SECTION 1

What does a generated picture show, and what does it prove?

> Distinguish a simplified generation mechanism from an observation of an event.

Your learning goals

I can explain the **mechanism** and terminology of image generation and **provenance**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall criteria and evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Prompt Evaluation: an attractive output must still be checked against explicit criteria. A generated picture also needs an honest record of its origin and of any factual claim made about it.

Your end goal

By the end, I can describe a toy **denoising process** and distinguish generated illustrations from **observation**s.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Prompt Evaluation. Give an example and a mistake that the idea helps you avoid.

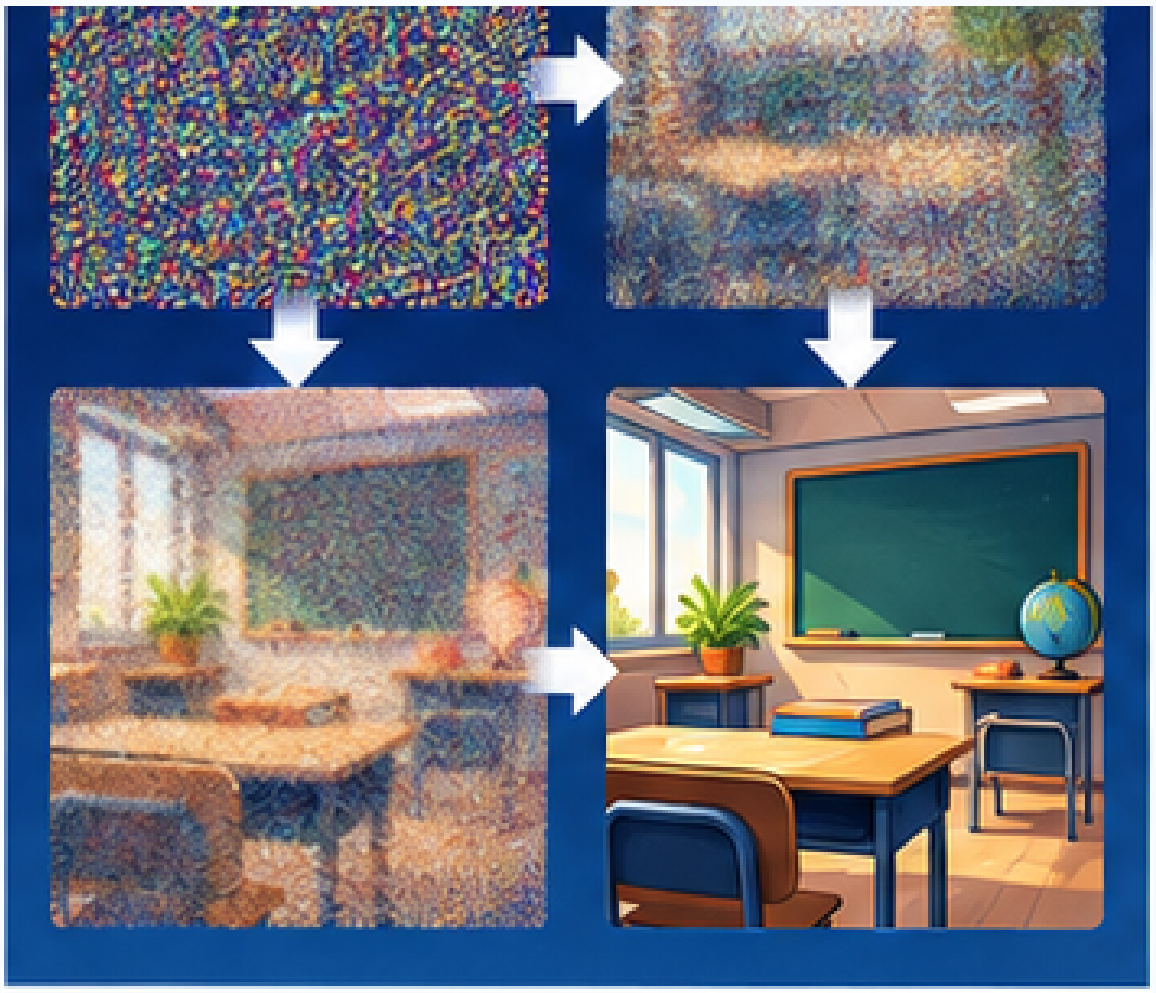
Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Variability. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 10: Image Generation Models

Image generation constructs visual output from learned patterns. Diffusion-style approaches use repeated updates; some operate on latent representations rather than directly on pixels.
</details>



CREATOR MISSION

Sequence a simplified **noise**-to-image card model.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

A generator constructs an image; a caption adds a claim

> A realistic appearance does not establish that an event occurred.

The generation-versus-observation distinction

An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Use supplied classroom illustrations within their scope

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

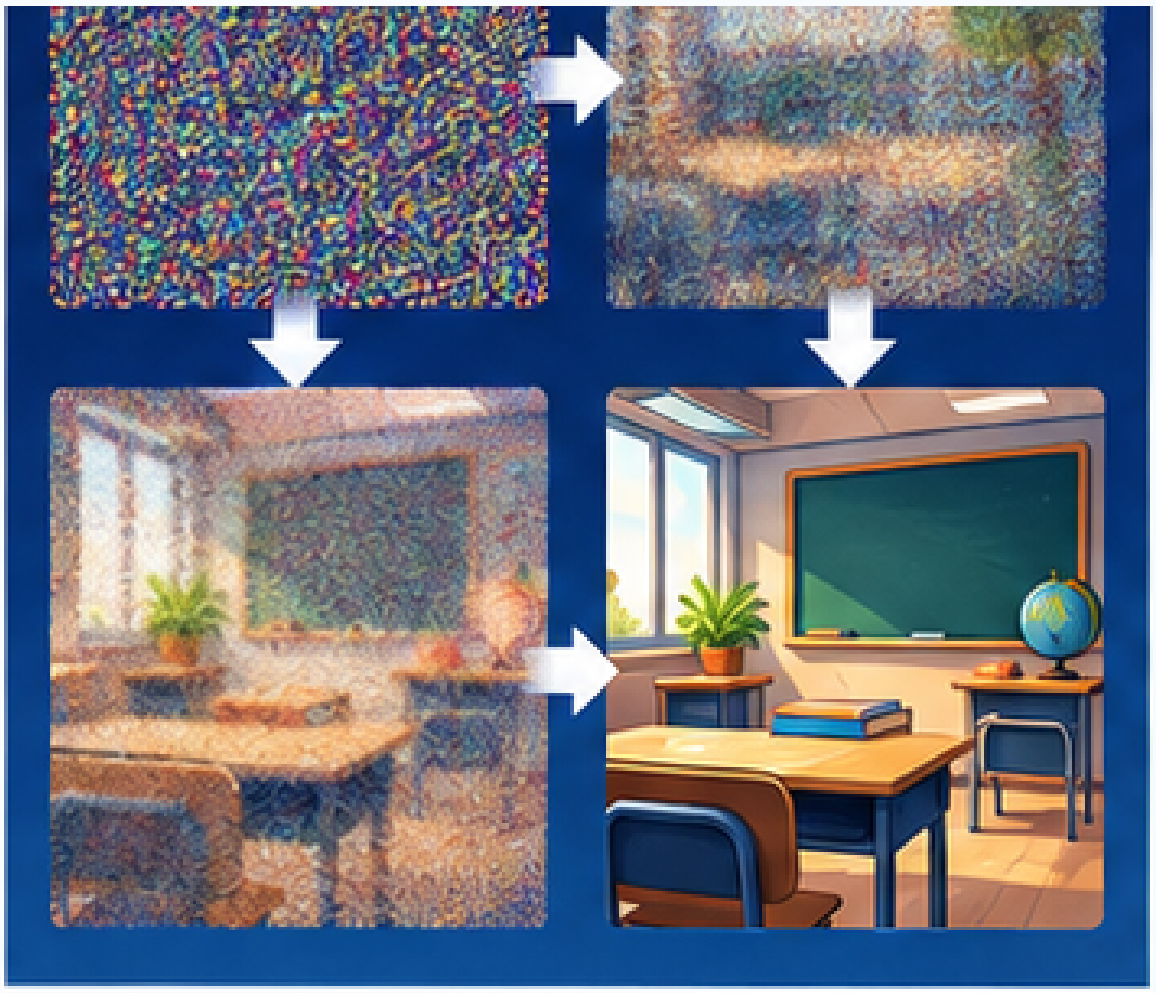
These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use image generation models to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Inspect visible requirements and the production record separately

> The constructed poster card makes the counted and labelled failures inspectable.

Record provenance without inventing unknown fields

Keep a **provenance** record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Read the complete constructed poster card

CONSTRUCTED POSTER CARD · INSPECT THE REQUIREMENTS

Required: exactly three circles labelled A, B, C from left to right.



Draft caption to audit: “Photograph of a real lesson.”

This is a fictional output card drawn for the lesson, not an actual generator run.

Count, legible labels and the authenticity claim must be checked separately.

Keep the criteria and evidence boundary

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the deciding visible evidence

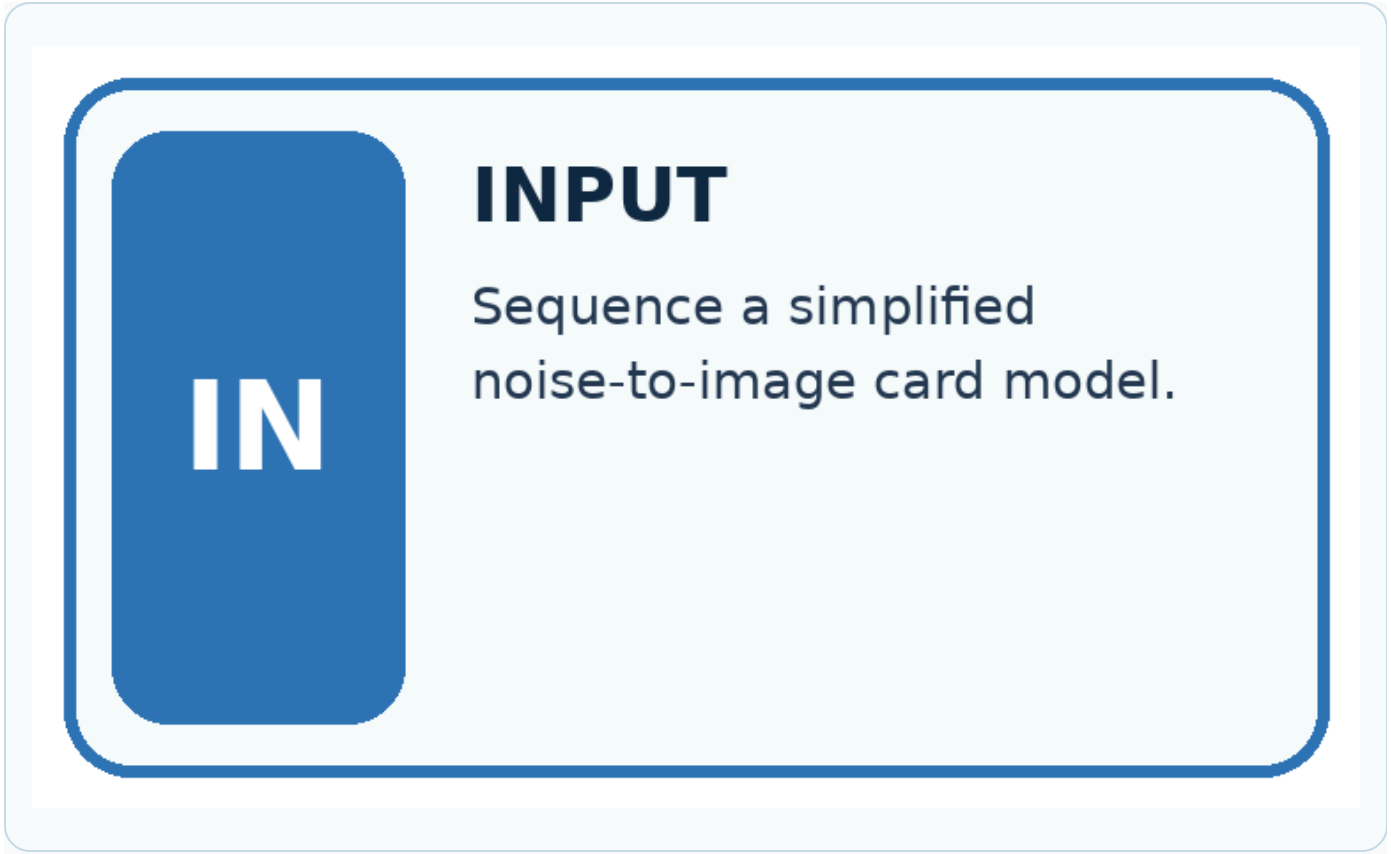
- Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

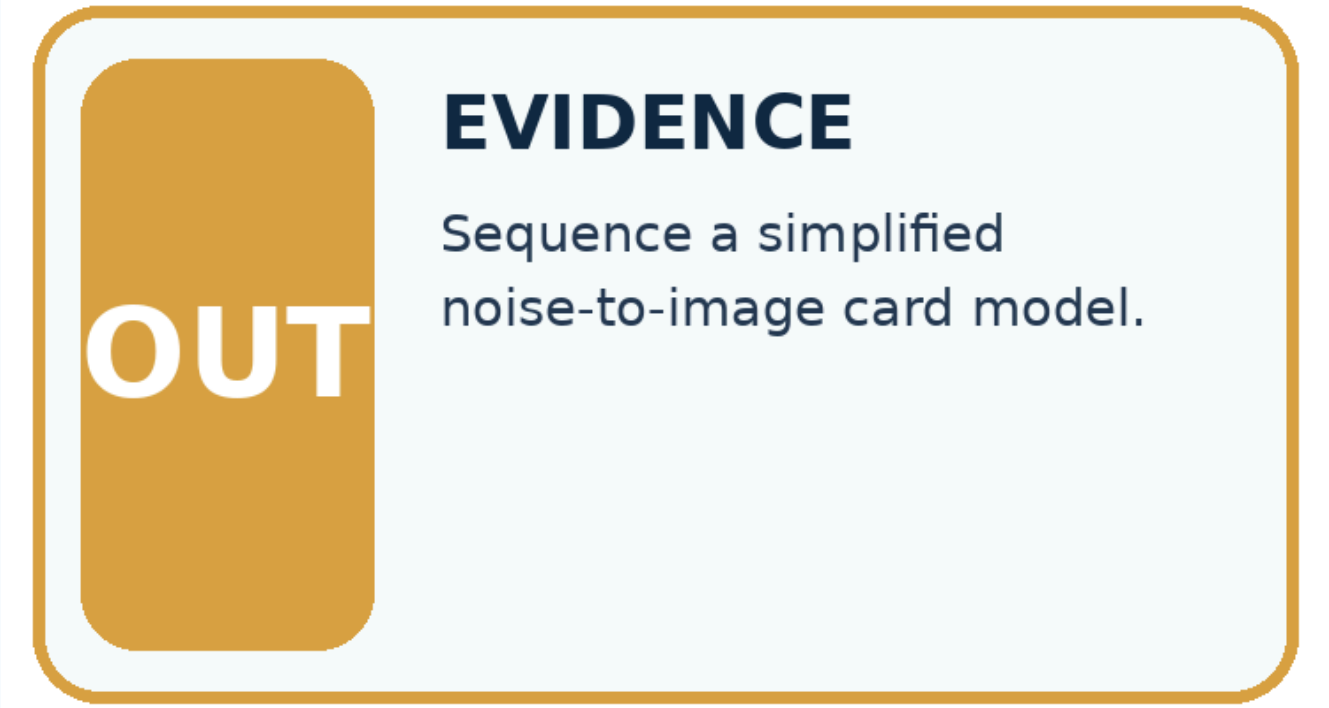
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Sequence a simplified noise-to-image card model.
↓
SYSTEM
Image generation iteratively constructs pixels guided by learned representations.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Image generation and provenance — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.	Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use.	It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

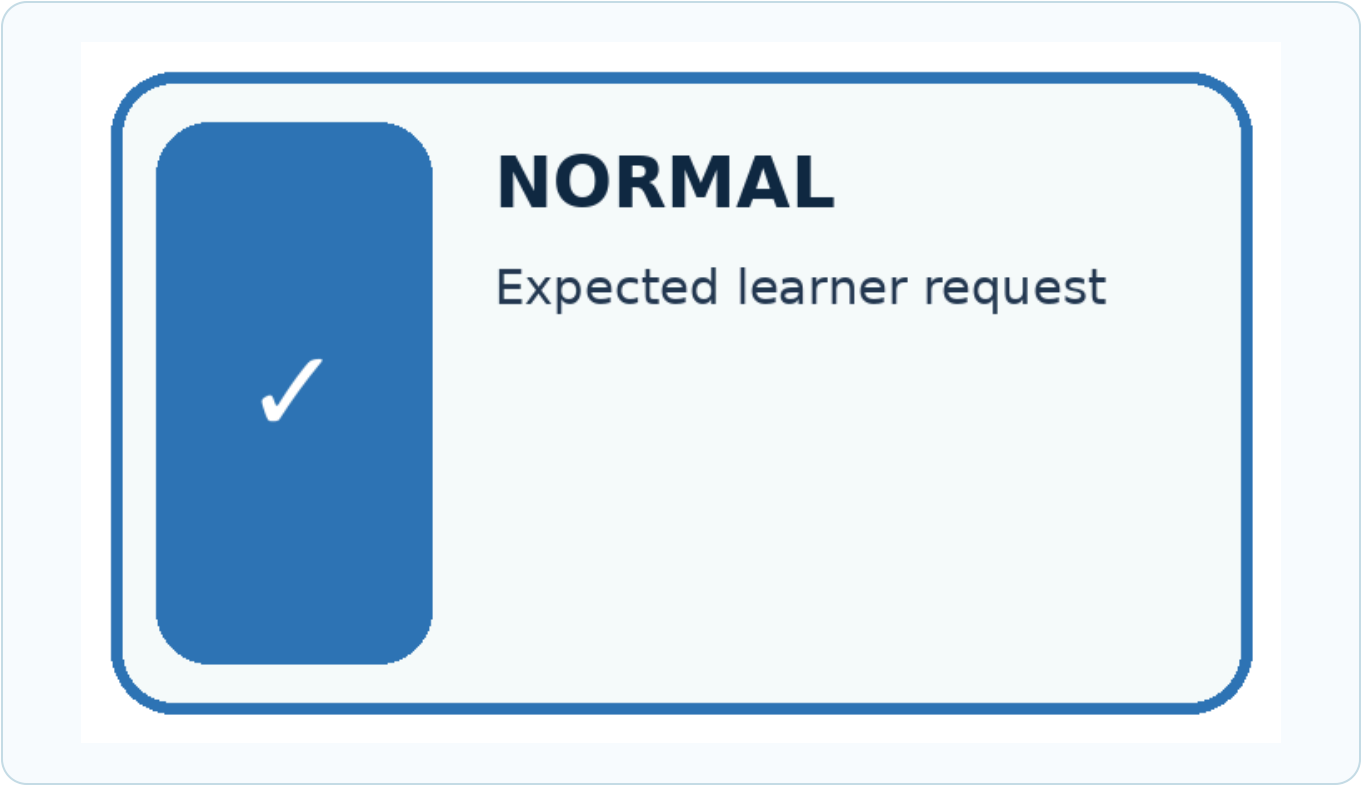
Connect the picture and the explanation

MAIN LESSON · SECTION 4

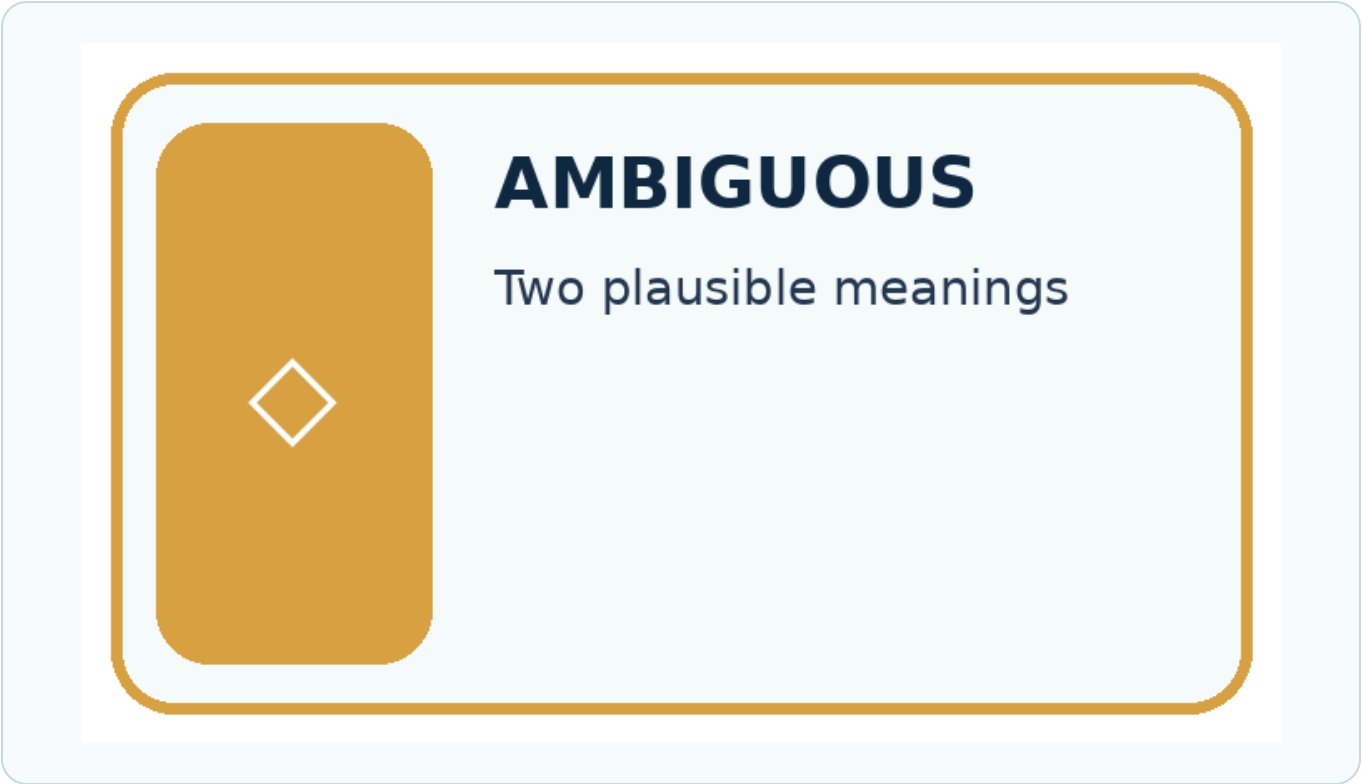
Compare a generated illustration with an authenticity claim

> Factual claims need evidence beyond the appearance of a picture.

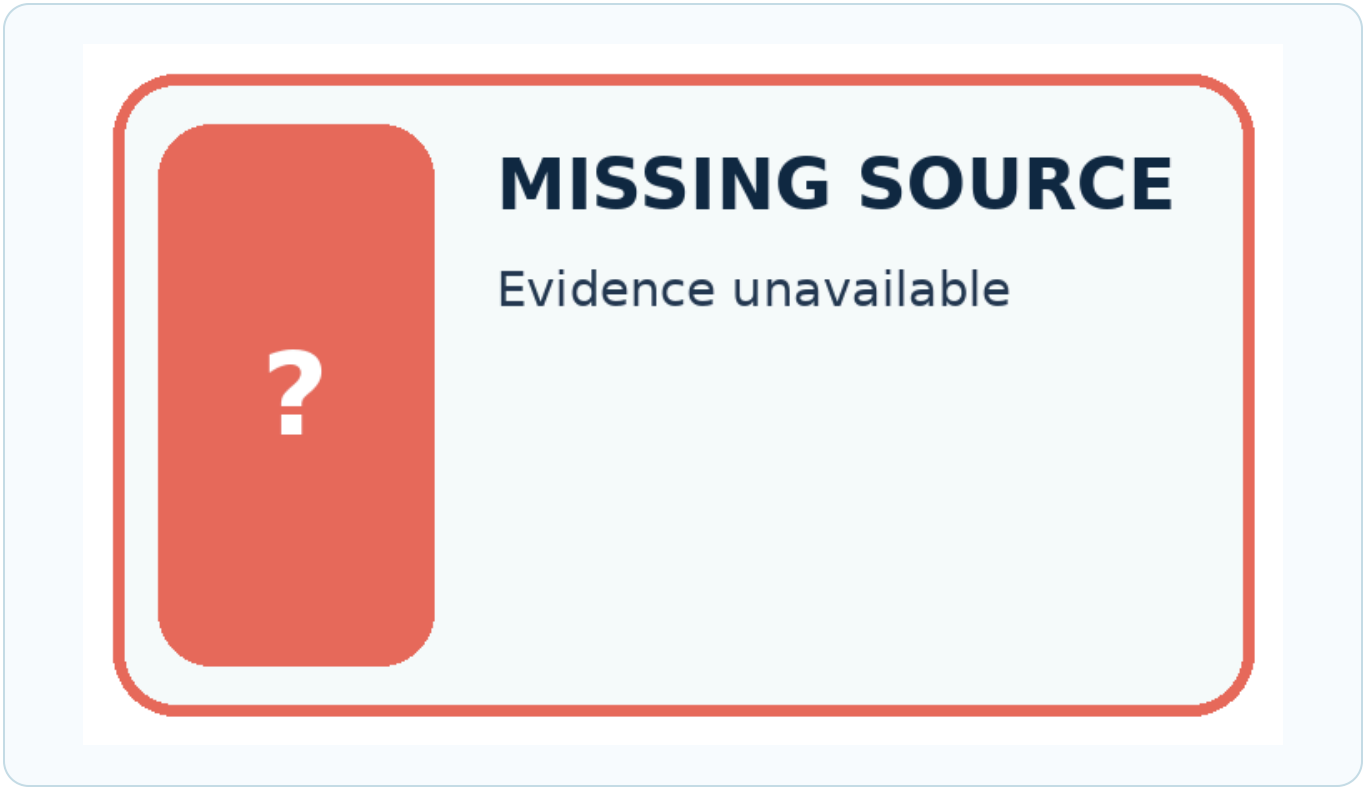
Four pictures with a purpose



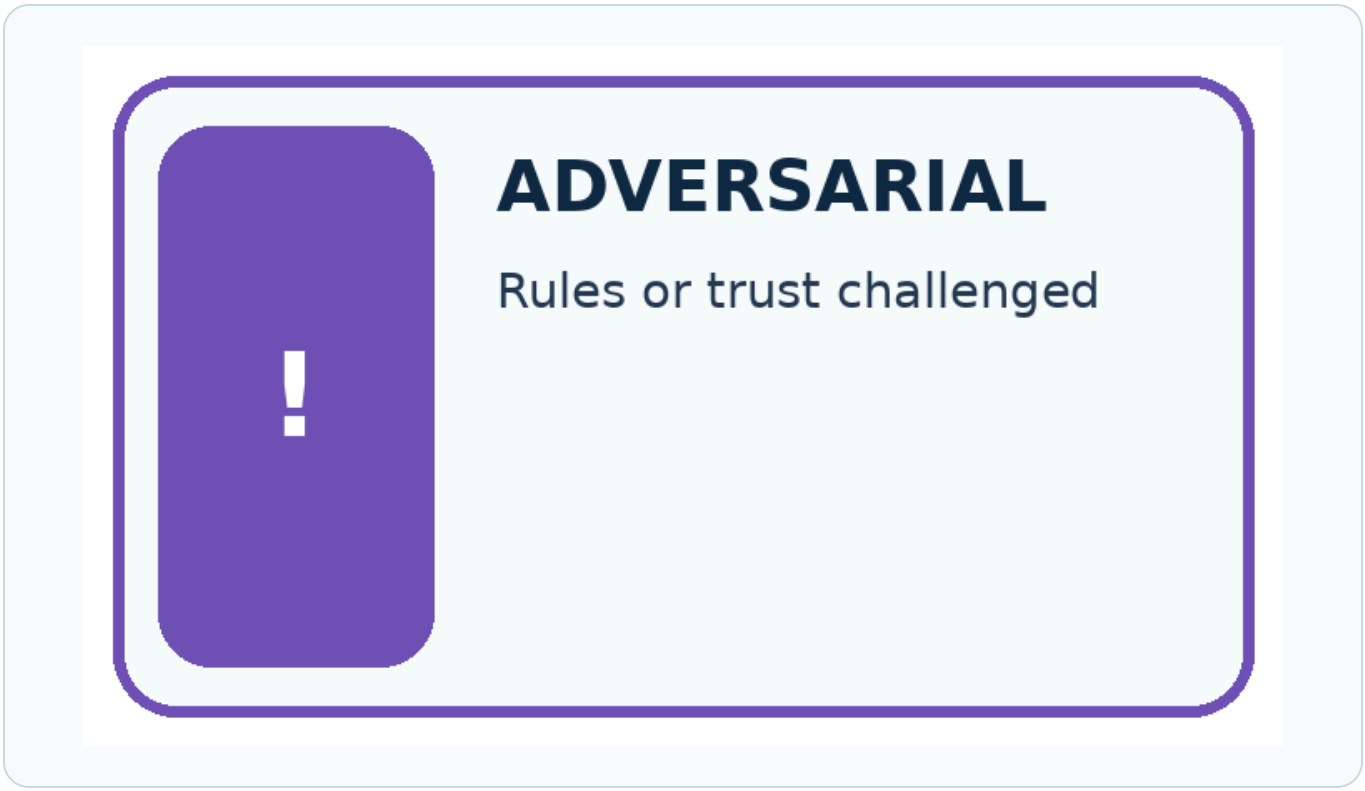
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Relabel the illustration as a photograph of an actual laboratory measurement.

<details>
<summary>Check after predicting</summary>

Prediction: That caption is unsupported.

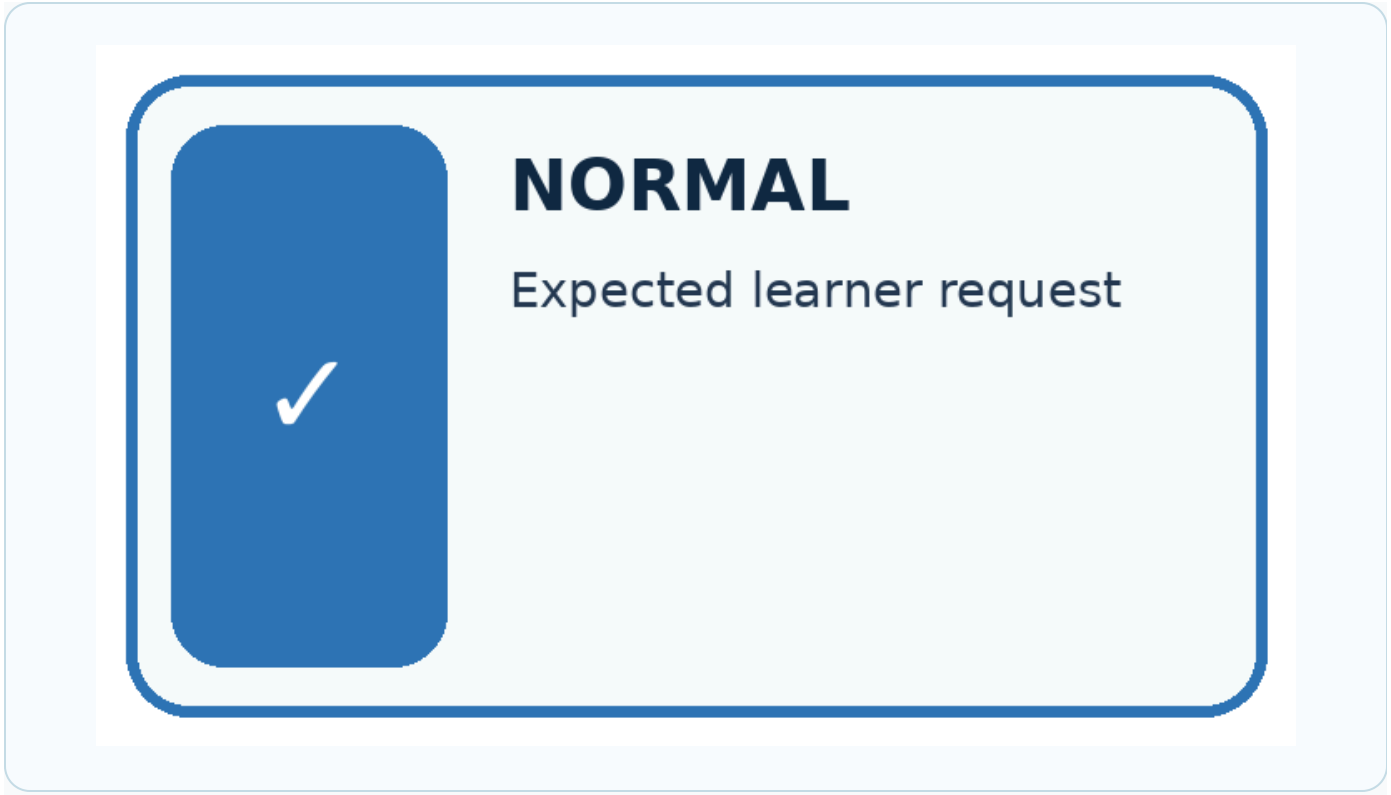
Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.
</details>

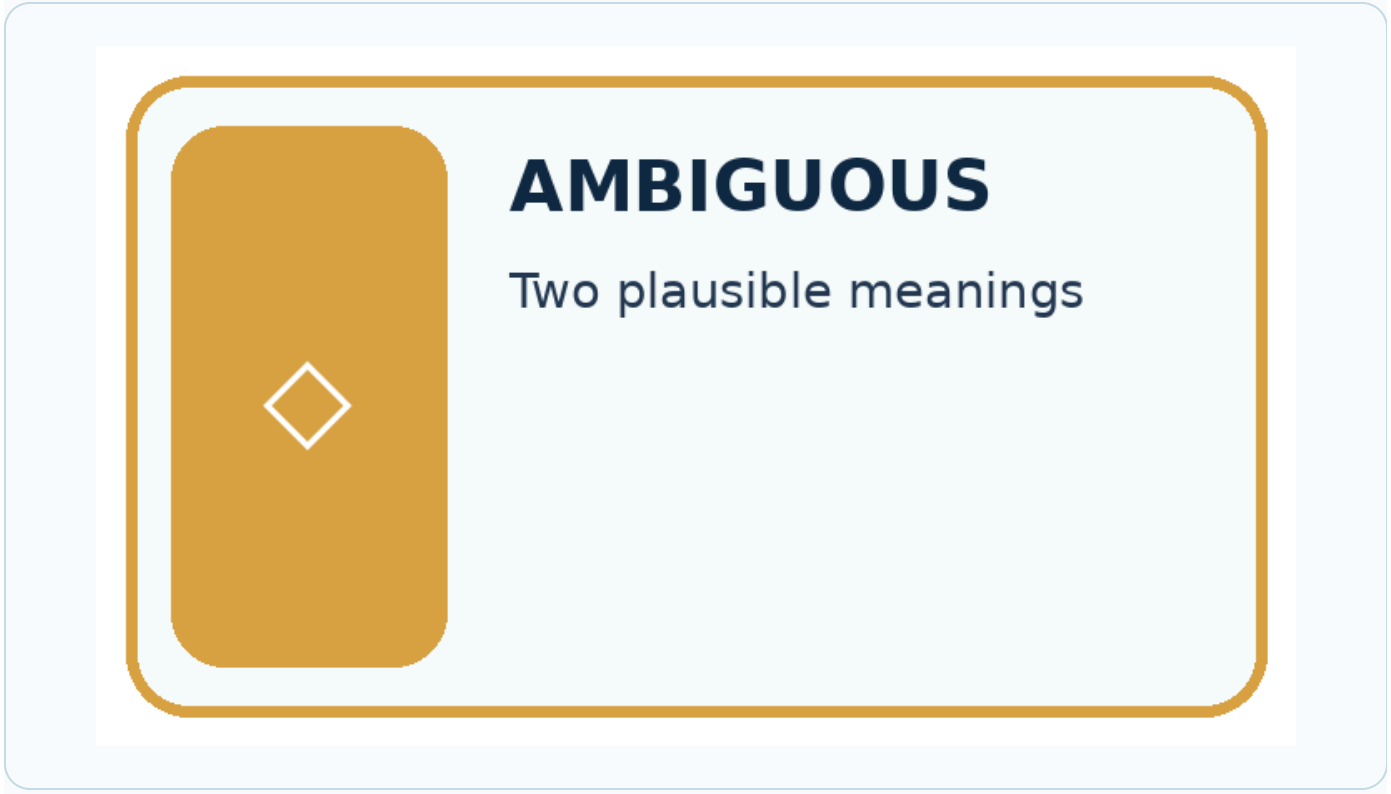
<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.
</details>





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace a supplied grid update without pretending to run diffusion

> The toy arrays illustrate a representation change, not a trained generator.

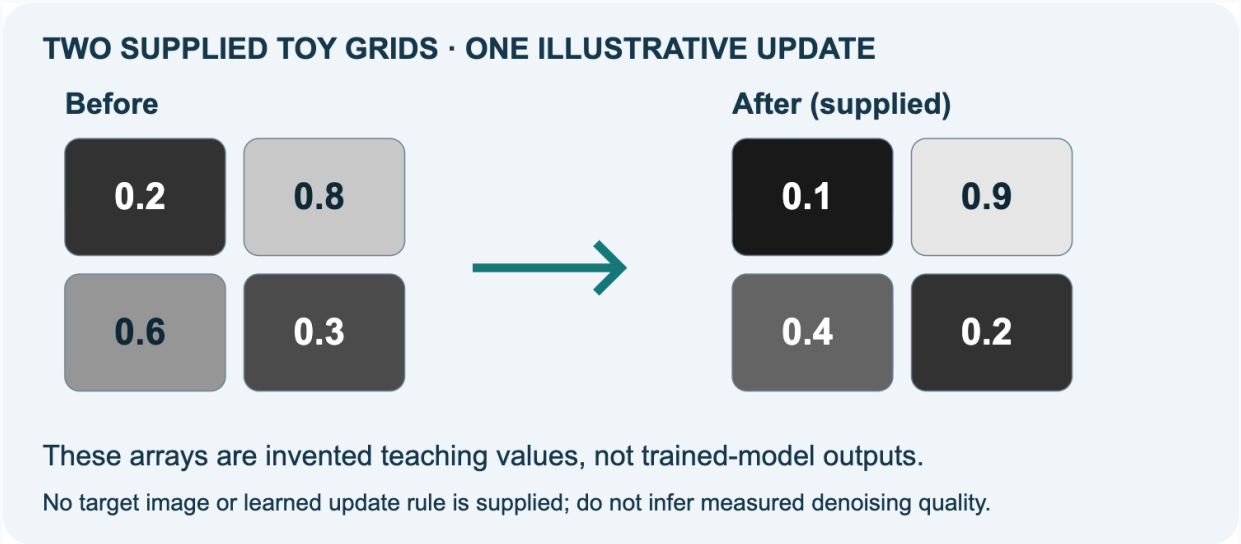
Read the two supplied arrays and their scope

A fictional lesson poster illustrates one family of image-generation approaches.

Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

Inspect both complete grids



Trace the imagined conditional update

AI ACADEMY / AI-3/10 · 02 · TRACE THE CASE

Follow the deciding steps

01 Noisy representation

Four toy numerical cells form the starting grid.

02 Condition on request

A text request guides the imagined learned denoising process.

03 Illustrative update

Use the supplied second grid to show one change in representation.

04 Check provenance

Caption it as a generated-process illustration, not a photographed experiment.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Noisy representation	Four toy numerical cells form the starting grid.
2. Condition on request	A text request guides the imagined learned denoising process.
3. Illustrative update	Use the supplied second grid to show one change in representation.
4. Check provenance	Caption it as a generated-process illustration, not a photographed experiment.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

State what the arrays cannot establish

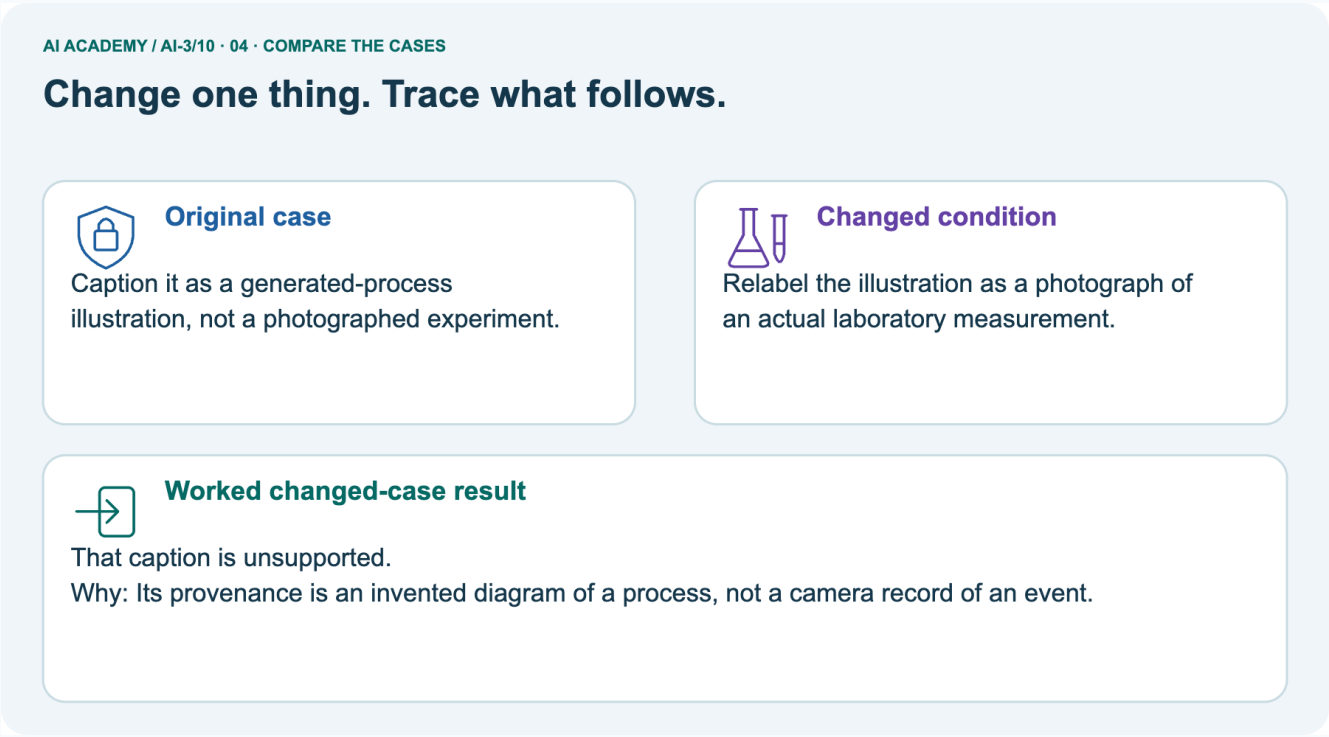
The second grid is supplied by the lesson, not calculated by a learned rule shown here. Per-cell changes are -0.1,+0.1,-0.2,-0.1 in row order. No clean target or loss function is supplied, so these values cannot establish that the update reduced error or produced a requested object. Four grayscale cells also do not demonstrate the separate poster's circle count.

Keep the numerical process illustration and poster-quality audit as distinct cases.

Predict changing only the authenticity caption

- Change one thing: Relabel the illustration as a photograph of an actual laboratory measurement. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the scope and architecture limit

- Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

- Read the labelled picture: What may these two grids explain, and what can they not establish about a real generator?

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

Worked evidence for Image Generation Models

STEP	EVIDENCE AND REASONING
1	In this simplified diffusion-style image generator, begin with random visual noise. Other image-generation methods use different procedures.
2	In this simplified account, use a learned estimate to update the noisy representation, with the request providing conditioning. Individual steps need not visibly improve.
3	Learned patterns influence shapes, textures, and relationships.
4	The result is generated, not retrieved evidence of a real event.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

1. In this simplified diffusion-style image generator, begin with random visual **noise**. Other image-generation methods use different procedures.

2. In this simplified account, use a learned estimate to update the noisy representation, with the request providing conditioning. Individual steps need not visibly improve.

3. Learned patterns influence shapes, textures, and relationships.

4. The result is generated, not retrieved **evidence** of a real event.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

A fictional poster requires exactly three labelled circles. Generated image shows four circles and one unreadable label. A caption calls it a photograph of a real lesson.

It fails count and label criteria; the photograph claim is unsupported. Record it as a generated fictional illustration and revise or reject before use.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Image Generation Models

The facts we will use

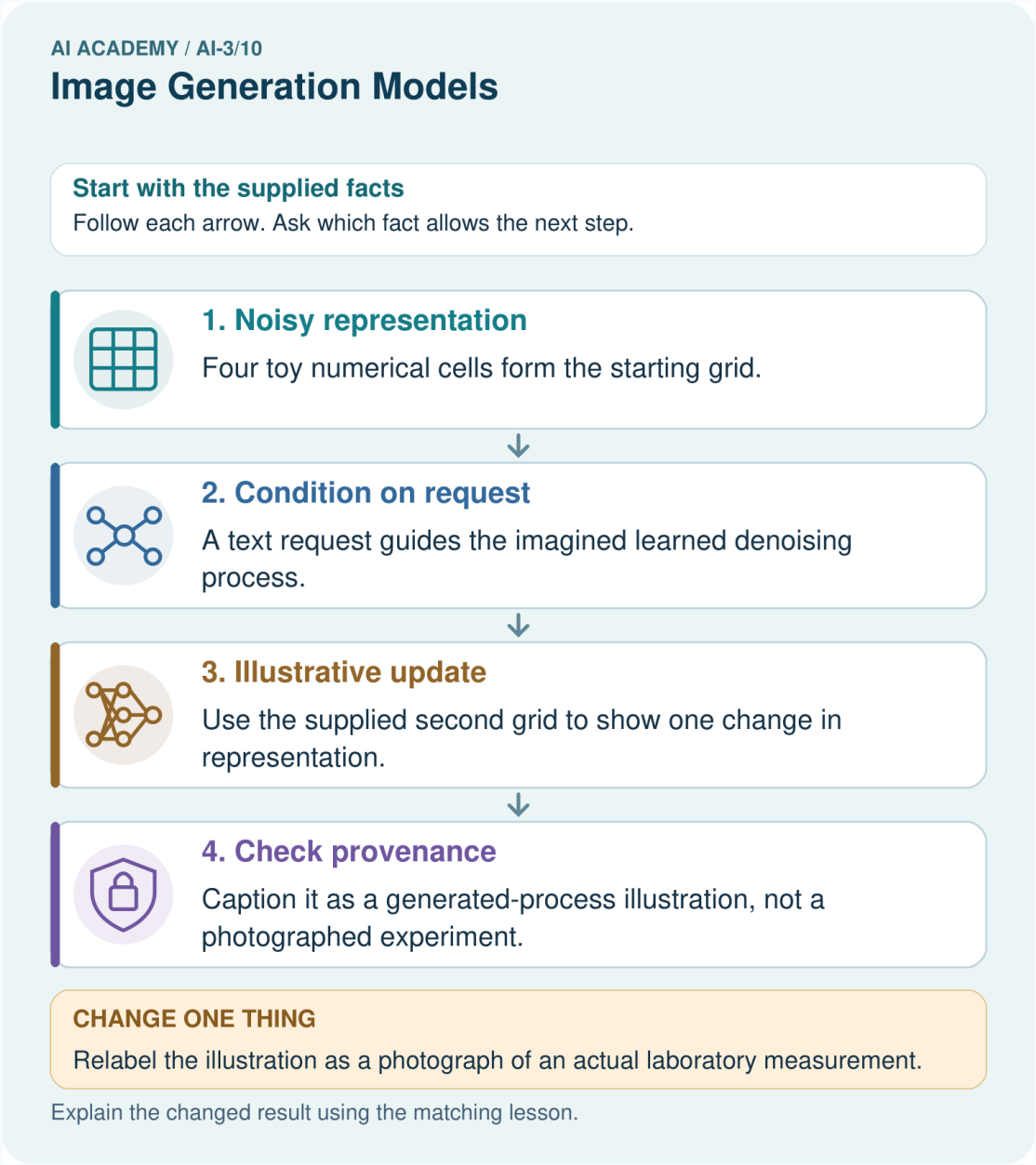
AI ACADEMY / AI-3/10

Illustrative grids, not executed model outputs

	Left	Right
Before top	0.2	0.8
Before bottom	0.6	0.3
After top	0.1	0.9
After bottom	0.4	0.2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find an authenticity claim the evidence cannot support

> Keep the teaching artifact and the claimed real event separate.

Repair the looks-real argument

Claim to repair: The after grid looks clearer, so it proves the model reconstructed a real scene. The arrays are invented teaching values and no model or target scene is supplied. The poster card is also constructed for this lesson. Both can explain an idea or expose a requirement failure; neither supplies a photograph, measured denoising accuracy or evidence that a depicted classroom event occurred.

Propose a provenance or criterion check

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: An image model searches a hidden folder and copies the closest picture. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

MAIN LESSON · SECTION 8

Practise the poster audit with less help

> Origin records and factual accuracy answer different questions.

State the provenance limit

Provenance tracks production history; it does not prove the content is accurate or establish every legal permission.

Complete poster W2

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Image generation and **provenance**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair an all-future-cases claim

- Use the worked case for Image generation and **provenance**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the supplied grids and their limit

- What may these two grids explain, and what can they not establish about a real generator?

Explain the picture

My explanation using a pictured step and the given facts

Reserve the independent diagram audit

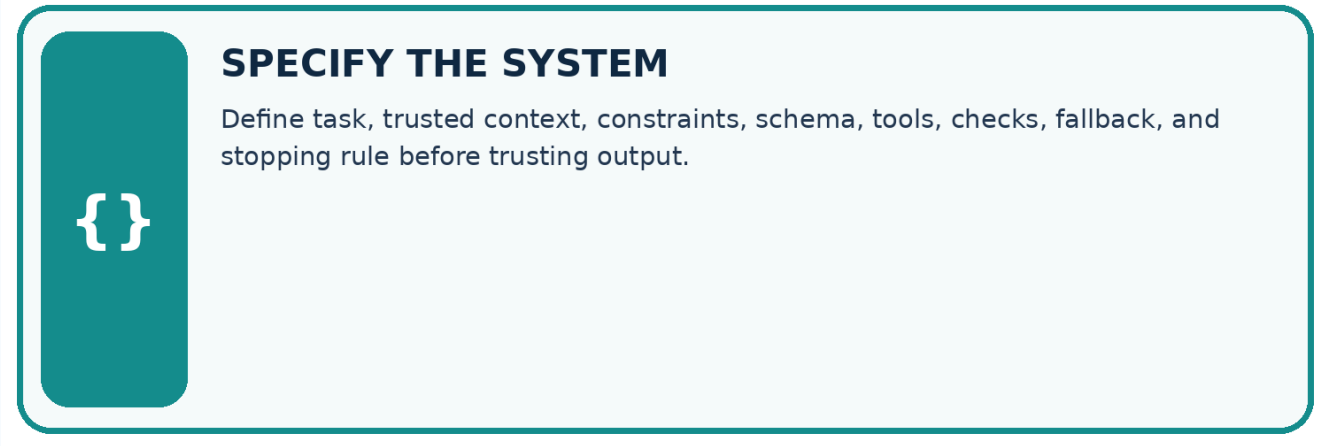
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

MAIN LESSON · SECTION 9

Use complete process cards and a separate visible poster audit

> The lesson supplies every card and the candidate image; no generation account is required.

The four process cards

Teaching card	Supplied content and role
K · Condition	Request: a fictional poster with exactly three readable circles A,B,C
L · Initial representation	Supplied2×2 grid[[0.2,0.8],[0.6,0.3]],declared noisy for this illustration
M · Illustrative update	Supplied second grid[[0.1,0.9],[0.4,0.2]]; no learned update function provided
N · Evidence boundary	No final generated poster follows from these arrays; record this as a process illustration

Trace the partial order and its evidence limit

1. Process-card investigation: K and L are inputs to the imagined update M, and N records its limits. The request and initial representation need not be placed in an invented single order relative to one another; both must be available for a conditioned update. Inspect the two full grids and describe changes without claiming measured improvement.

   This is a conceptual diagram and supplied-array trace, not an implemented diffusion algorithm or observed generator run.

Audit the separate poster and honest production record

1. Separate poster audit: use the constructed four-circle card and its explicit three-circle A/B/C requirement. Count circles, inspect label completeness/legibility and critique the draft photograph caption independently. Keep a provenance record stating native classroom drawing, no model run, no generator/version or measured generation date available, and the intended teaching use.

   Unknown generator fields must remain not supplied. Propose one change to the specification or drawing and its expected check, but do not invent a newly generated output. Exact counts and labels can be constructed with a vector/diagram tool and still require inspection. Keep the separate two-arrow assessment withheld.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

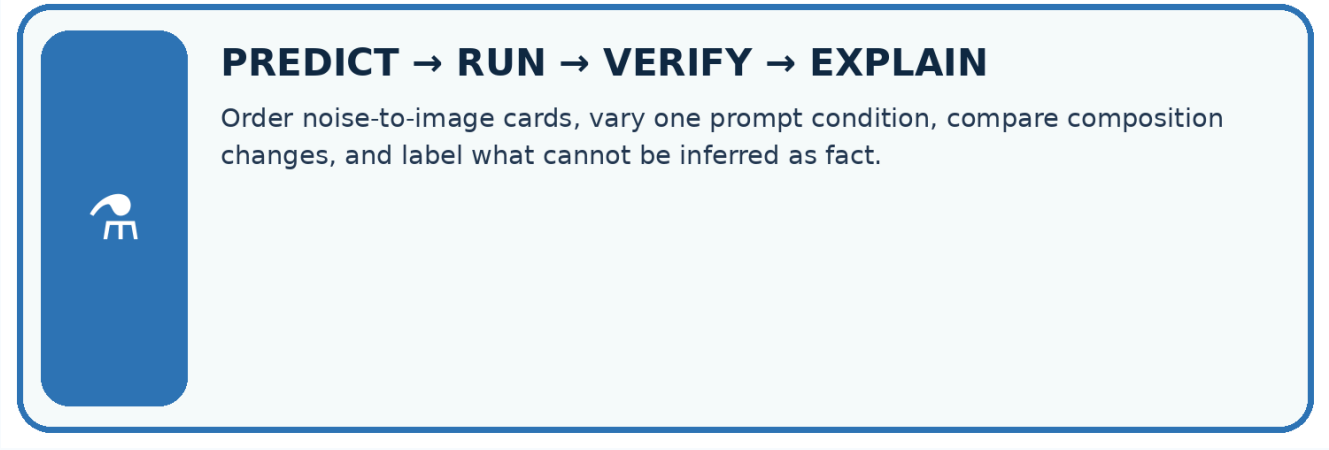
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the supplied process cards, numerical grids and separate constructed poster card. Trace inputs and illustrative update, then check visible counts/labels and provenance. No real generator is run and no final image or denoising-quality measurement follows from the arrays.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: audit the fresh two-arrow diagram record

> A correct count cannot repair swapped meanings or an unsupported observation claim.

Independent record and scope

A generated diagram requires two arrows and labels **input**/**output**. It has two arrows but swapped labels. Metadata records generation, yet the caption says experimental **observation**. Audit.

Audit the supplied description of the independent diagram. No actual image or camera record is provided, so do not invent geometry or an observed experiment. Treat arrow count, label meaning and the authenticity caption as separate requirements.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

An image model draws a plant with seven roots after a request for exactly four. What does this show, and where should an exact count diagram be made?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Treat generated diagrams as illustrations; pair factual claims with approved sources.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

An image generator produces a convincing picture of a school greenhouse on Mars. A student calls it photographic **evidence**. Explain what the image can and cannot establish.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's diagrams honest and checkable

> Pair each educational label or factual claim with the evidence it needs.

This week's concrete contribution

For the Verified Study Buddy, label diagrams as constructed or generated illustrations with the known production steps and edits. Check counts, labels, spatial relations and factual explanations against the actual requirement and approved sources. Keep unknown tool/version fields unknown.

A production record supports the origin claim it documents; it does not verify every depicted fact or create an observation of a fictional scene.

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels.

Keep the checkpoint and human choice

THIS WEEK'S CHECKPOINT Treat generated diagrams as illustrations; pair factual claims with approved sources.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

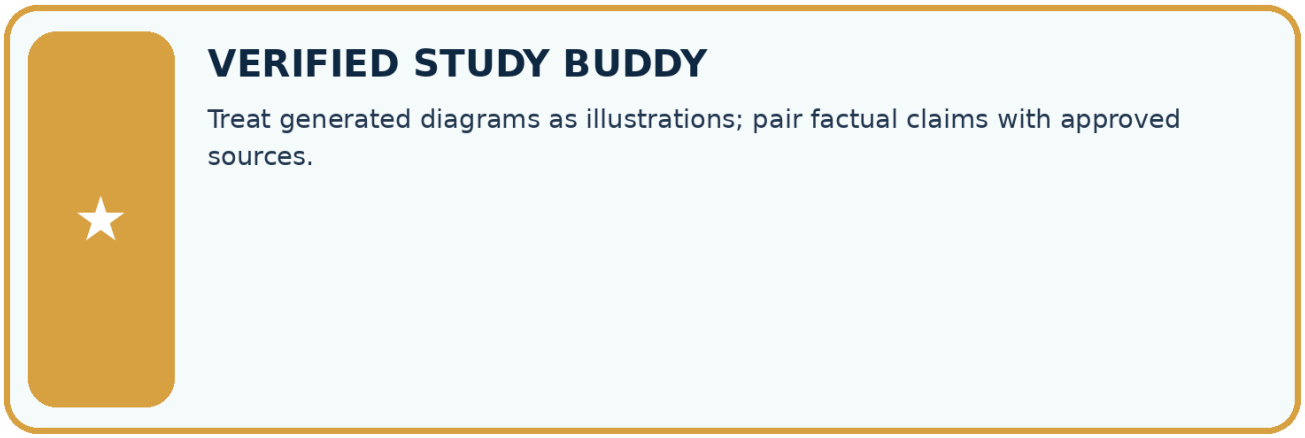
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain the process illustration and its evidence boundary

> A plausible visual is not proof that its scene occurred.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of image generation and **provenance**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
latent representation	An internal representation learned or constructed by a model rather than directly supplied as the original observation.	
noise	Variation or interference that obscures the information relevant to a task.	
denoising	Reducing noise in a representation; some generative models learn repeated denoising steps to create outputs.	
conditioning	Making a model's output depend on supplied information, such as a prompt or image.	

Terms for the connected explanation

Word	Meaning
Provenance	Recorded origin and transformation history

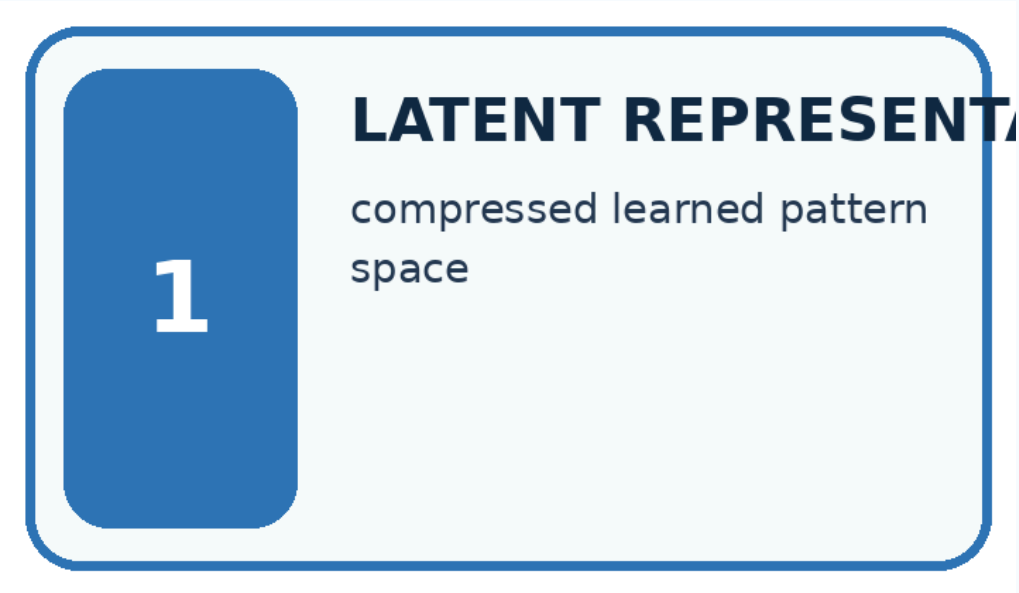
Word	Meaning
Visual criterion	A stated requirement checked against an image
Authenticity claim	A claim about the source or reality of depicted content

▼ Earlier wording, alternatives and picture transcripts

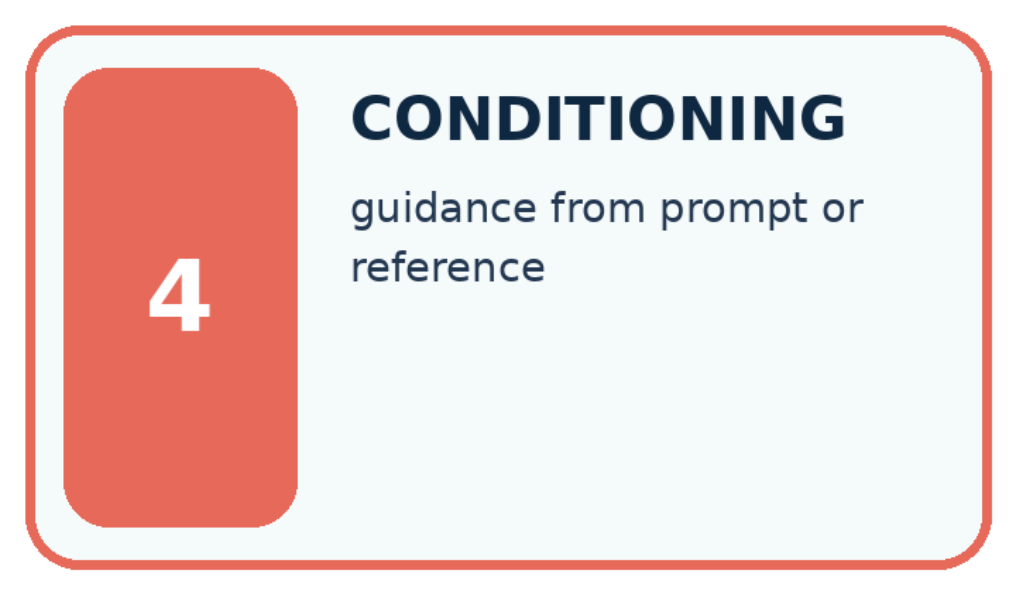
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
NOISE
random starting signal
3
DENOISING
iterative prediction toward structure



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

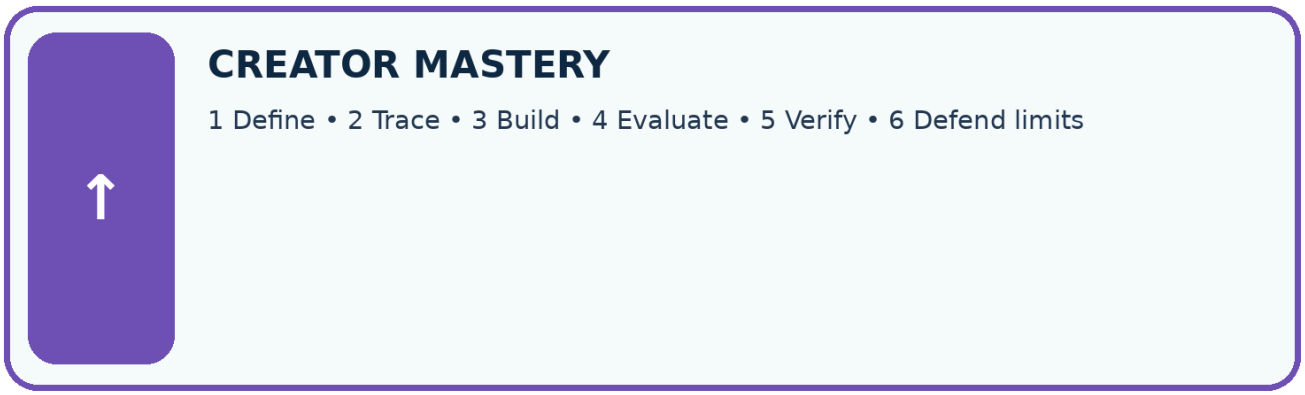
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Audit a new diagram independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

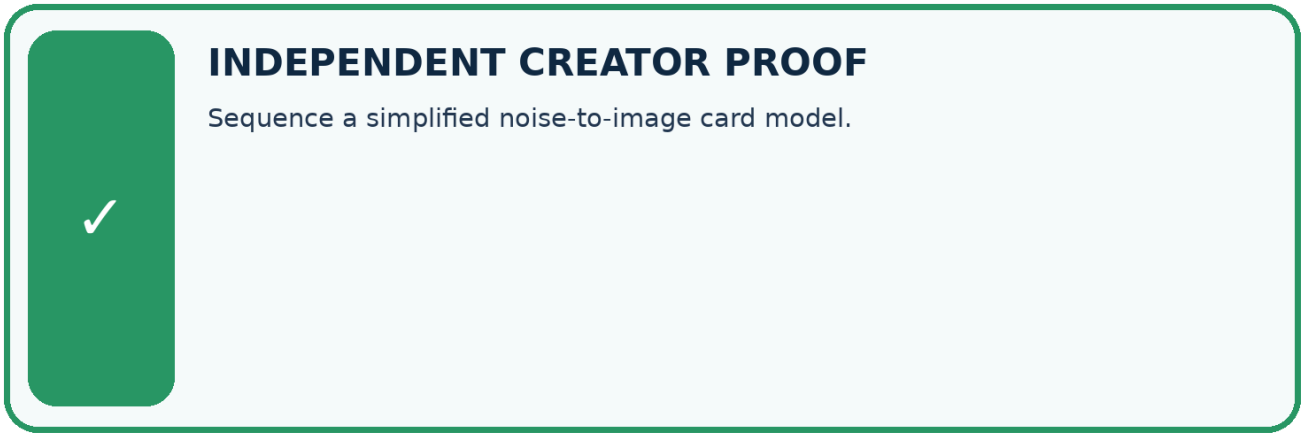
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Audit a new diagram independently is the selected independent case.



EXPLANATION CHECK Keep count, label meaning, known origin and factual authenticity as separate checks.

Write or draw your reasoning and evidence.

▼ **Extra practice: A generated image does not document an event**

REFERENCE · SECTION 13

Extra practice: A generated image does not document an event

> Try the question before opening its worked explanation.

Read the supplied case and question

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

Evidence or stage	Value or result
Brief	Bicycle beside a tree
Model	Synthesises visual patterns
Image	Fictional scene
Check	Content and intended use

▼ Earlier wording, alternatives and picture transcripts

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

A generated image does not document an event

A generator creates a realistic fictional bicycle beside a tree.

How does this differ from event photography?

It synthesises an image from learned patterns and instructions; it does not document that event. Many systems de**noise** iteratively, but architectures vary.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Explain the mechanism and terminology of image generation and provenance	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional lesson poster illustrates one family of image-generation approaches.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional lesson poster illustrates one family of image-generation approaches.
- Which labelled step matches this detail? Caption it as a generated-**process** illustration, not a photographed experiment.
- 1. Noisy representation
- 2. Condition on request
- 3. Check **provenance**
- 4. Illustrative update
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check **provenance** Caption it as a generated-**process** illustration, not a photographed experiment.

Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Changed-case question

- Changed-case practice: Relabel the illustration as a photograph of an actual laboratory measurement. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. That caption is unsupported.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That caption is unsupported. Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Repair and save your work

Review Prompt Evaluation (ai-3/9). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Why is B's selection defensible here without calling it universally superior?.

Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Sampling and Variability (ai-3/4). Goal: Sample from a distribution using explicit cumulative intervals. Short practice: Does T=0.5 make B impossible, and does choosing A guarantee factual accuracy?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

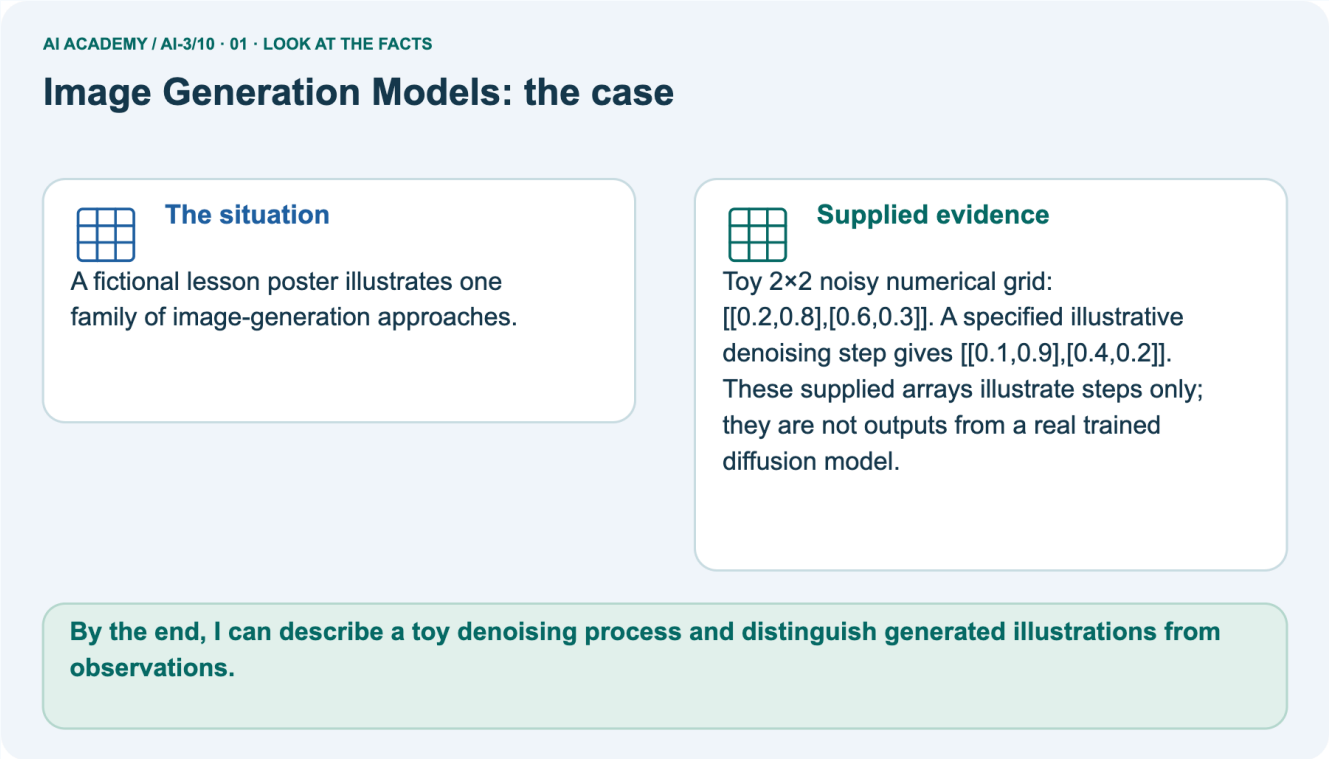
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Image Generation Models: the case



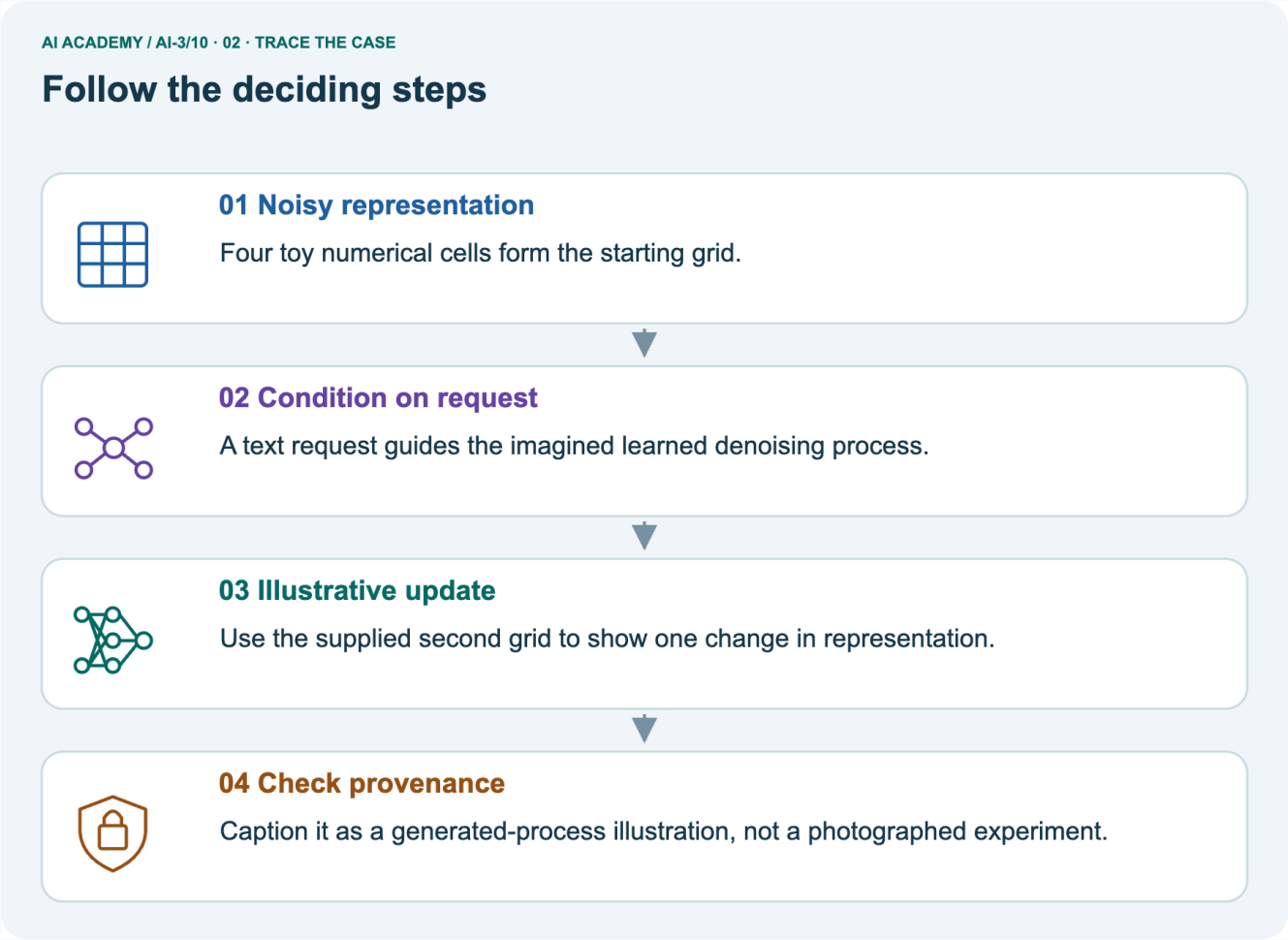
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional lesson poster illustrates one family of image-generation approaches. Supplied facts: Toy 2×2 noisy numerical grid: [[0.2,0.8],[0.6,0.3]]. A specified illustrative **denoising** step gives [[0.1,0.9],[0.4,0.2]].

These supplied arrays illustrate steps only; they are not **output**s from a real trained diffusion model.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

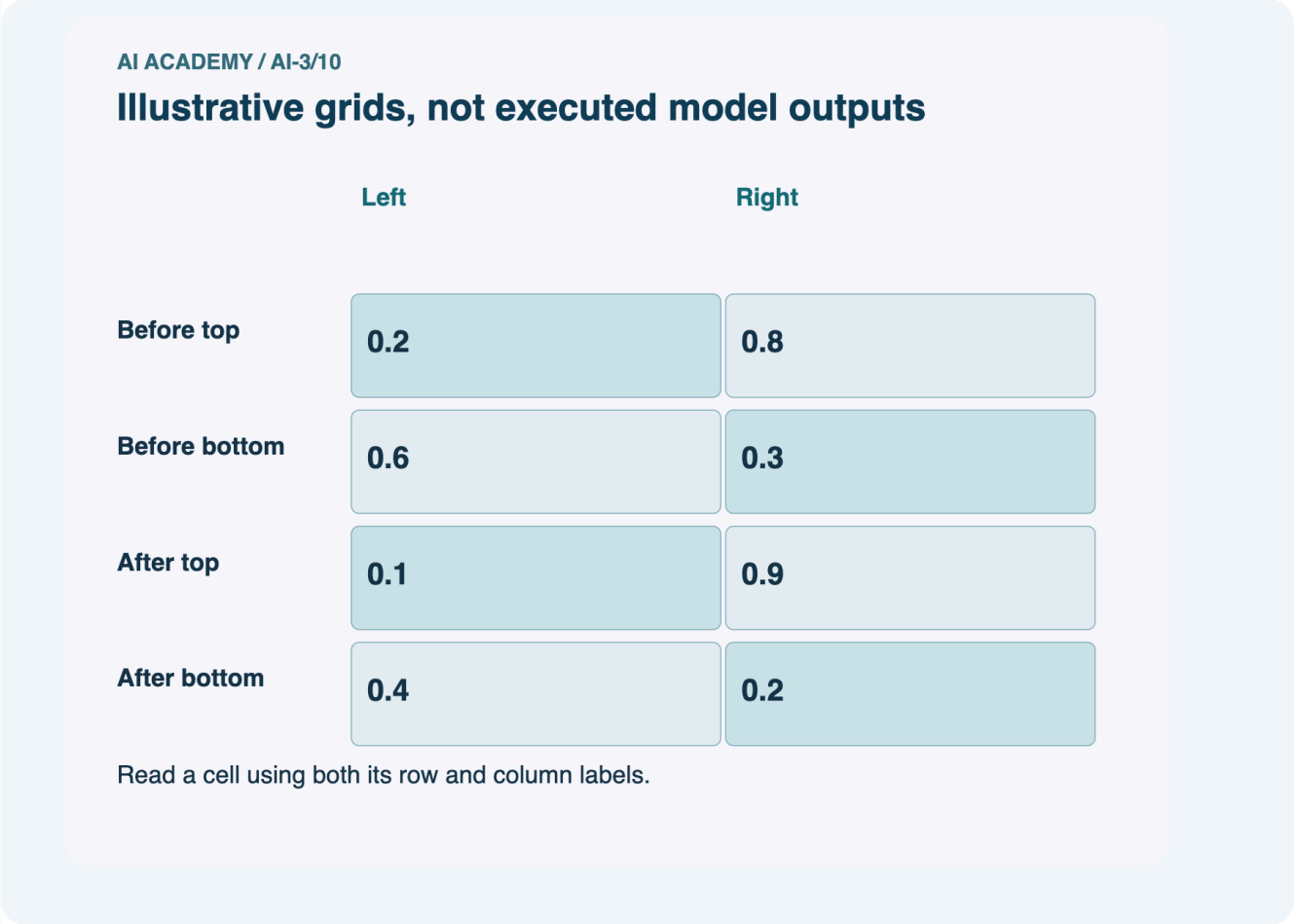


Follow the labelled picture with the matching explanation. Open the vector diagram

Noisy representation: Four toy numerical cells form the starting grid. → Condition on request: A text request guides the imagined learned **denoising process**. → Illustrative update: Use the supplied second grid to show one change in representation. → Check **provenance**: Caption it as a generated-**process** illustration, not a photographed experiment.

Point to each numbered stage and explain which recorded **input** it uses.

03. Illustrative grids, not executed model outputs

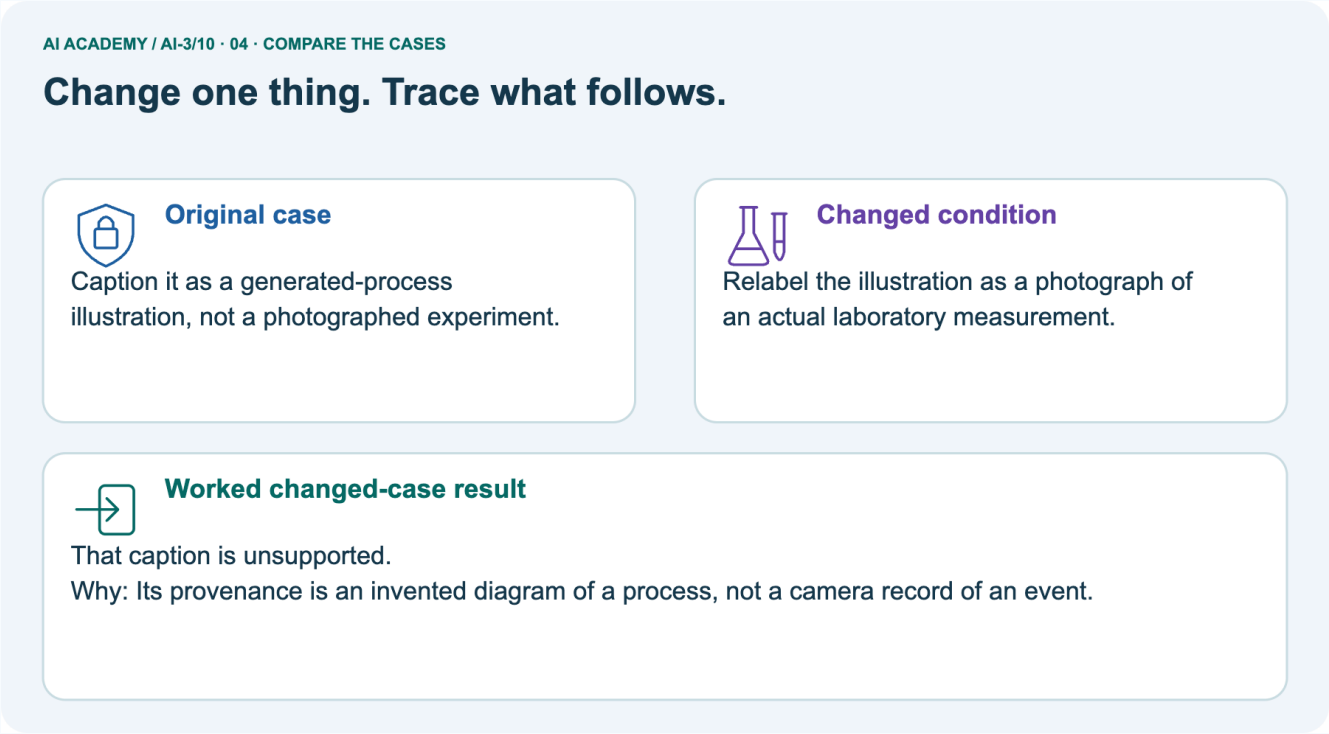


Follow the labelled picture with the matching explanation. Open the vector diagram

Illustrative grids, not executed model **output**s. Before top: Left = 0.2, Right = 0.8. Before bottom: Left = 0.6, Right = 0.3. After top: Left = 0.1, Right = 0.9. After bottom: Left = 0.4, Right = 0.2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



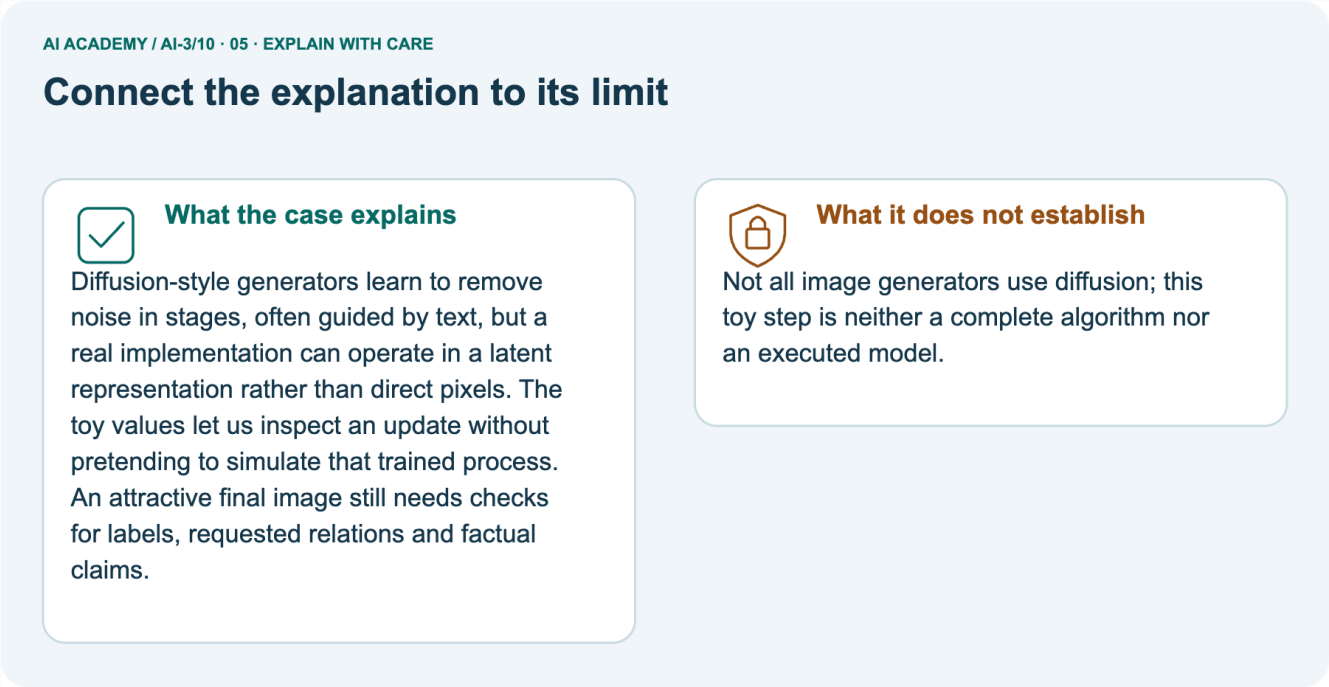
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Caption it as a generated-**process** illustration, not a photographed experiment. Changed condition: Relabel the illustration as a photograph of an actual laboratory measurement.

Expected result: That caption is unsupported. Reason: Its **provenance** is an invented diagram of a **process**, not a camera record of an event.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

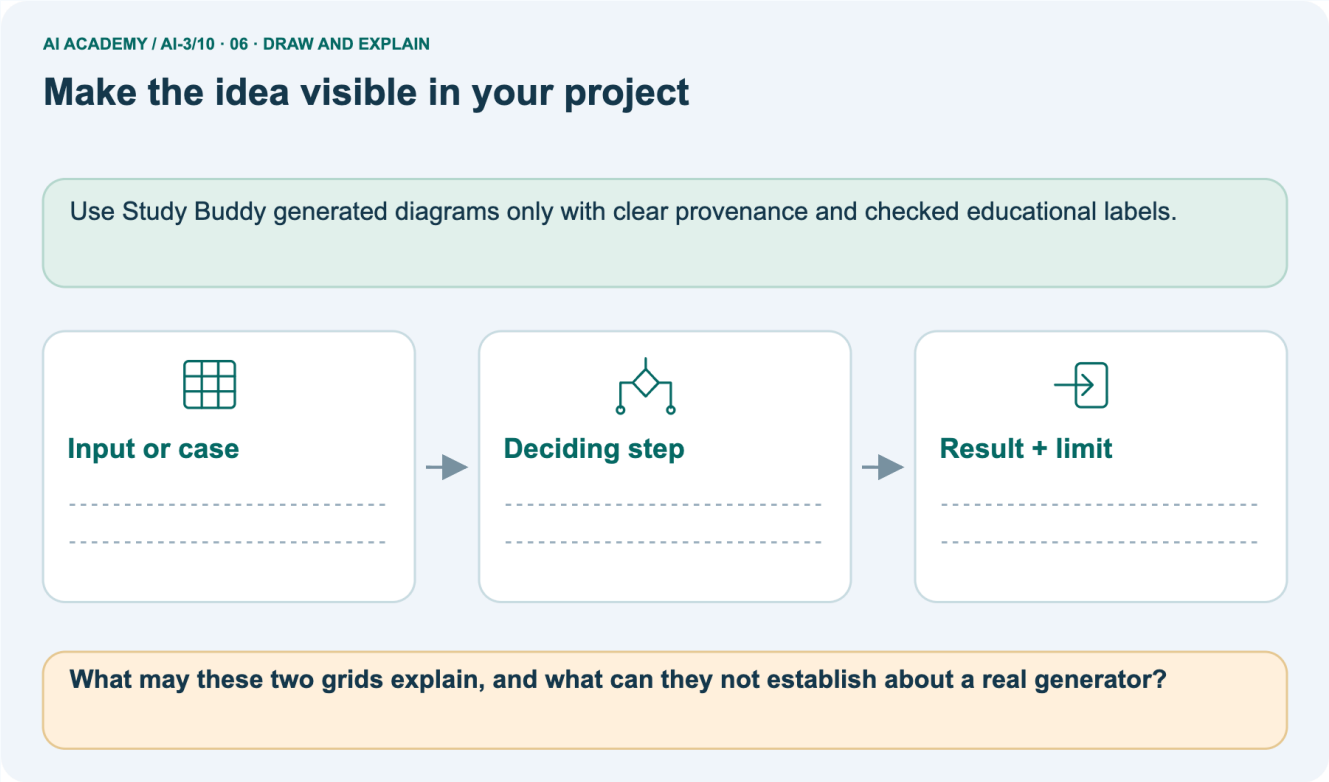
Diffusion-style generators learn to remove **noise** in stages, often guided by text, but a real implementation can operate in a **latent representation** rather than direct pixels.

The toy values let us inspect an update without pretending to simulate that trained **process**. An attractive final image still needs checks for labels, requested relations and factual claims.

Limit: Not all image generators use diffusion; this toy step is neither a complete algorithm nor an executed model.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy generated diagrams only with clear **provenance** and checked educational labels. Draw your **input**, deciding step, result and limit.

Supported question: What may these two grids explain, and what can they not establish about a real generator?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Image Generation Models: the case

02. Follow the deciding steps

03. Illustrative grids, not executed model outputs

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.